RAMCO INSTITUTE OF TECHNOLOGY
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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering
Academic Year 2022 – 2023 (Odd Semester)

Degree, Semester & Branch: I Semester B.E. CSE ‘A’
Course Code & Title: GE3151 Problem Solving and Python Programming
Name of the Faculty member (s): Mrs.B.Vijayalakshmi, AP (SG)/CSE

Innovative Practice Description

- **Unit / Topic: Unit II / Precedence of Operators**
- **Course Outcome:** CO2
- **Topic Learning Outcome:** TLO5
- **Activity Chosen: Think Pair Share**
- **Justification:**
  The understanding of different types of operators and its precedence helps the students to find solution to a problem and make them to apply the operator in a correct order to get exact solutions. Think pair share activity helps them to recollect and think operator precedence.
- **Time Allotted for the Activity: 10 Minutes**
- **Details of the Implementation:**
  - A Collaborative, active learning strategy, in which students work on a problem posed by instructor in a worksheet, first individually (Think), Write in a paper, then work in pairs or groups (Pair) to solve the problem, and finally share their solution with the entire class (Share).
  - **Think – Write-** Students were provided with worksheet of questions on operators. The student has to think about the precedence of operators and find the solutions individually and write the answer in a paper.
  - **Pair-** After writing the answers, the students were allowed to discuss with neighbour and justify the resultant answer to each other which is shown in Fig.1. The answers written by the students are shown in Fig.2.
  - **Share -** Once the conclusion is made between the pair, then it is to be shared among the whole class by any of the student from the team.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO2	3	2	2	2	1	1	1	1	3	2	3

(1 – Low 2 – Moderate 3 – High)

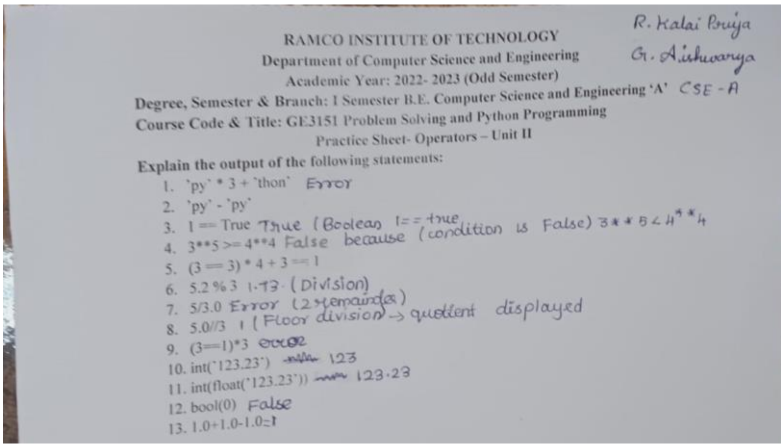
- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1	PSO2	PSO3
	3	2	2	1	1	3	1	1

Justification for correlation	The basic knowledge about operator is needed	The students analyze the given questions and apply the operator precedence rule.	The students find ways to solve the given problems.	The students coordinate and demonstrate the activity as a team.	The students communicate the solution to others.	The concept is used in developing the software	The concept is used in security applications	The concept is used in solving problems in AI

- **Images / Screenshot of the practice:**

Fig 1: Students Involvement



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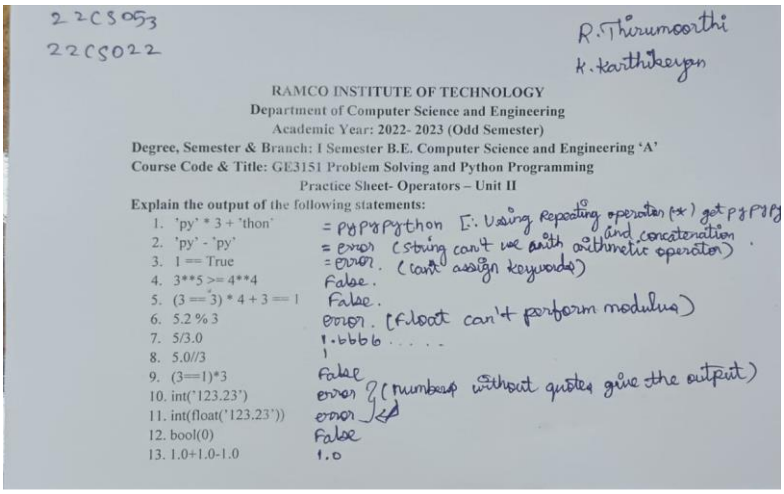
Course Code & Title: GE3151 Problem Solving and Python Programming
Practice Sheet- Operators – Unit II

Explain the output of the following statements:
1. 'py' * 3 + 'thon' Error
2. 'py' - 'py'
3. 1 == True True (Boolean 1==true condition is False) 3**5<4**4
4. 3**5 >= 4**4 False because
5. (3 == 3) * 4 + 3 == 1
6. 5.2 % 3 1.73 (Division)
7. 5/3.0 Error (2 remainder)
8. 5.0//3 1 (Floor division) → quotient displayed
9. (3==1)*3 error
10. int('123.23') 123
11. int(float('123.23')) 123.23
12. bool(0) False
13. 1.0+1.0-1.0=1



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Practice Sheet- Operators – Unit II

Explain the output of the following statements:
1. 'py' * 3 + 'thon' = pypypython [∵ Using Repeating operator (*) get pypypy and concatenation
2. 'py' - 'py' = error (string can't use with arithmetic operator)
3. 1 == True = error. (can't assign keywords)
4. 3**5 >= 4**4 False.
5. (3 == 3) * 4 + 3 == 1 False.
6. 5.2 % 3 error. (float can't perform modulus)
7. 5/3.0 1.6666 . . . . .
8. 5.0//3 1
9. (3==1)*3 False
10. int('123.23') error (numbers without quotes give the output)
11. int(float('123.23')) error
12. bool(0) False
13. 1.0+1.0-1.0 1.0

Fig. 2: Sample Sheet

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - Some of the students easily solve the problems and share their ideas with others.
    - Some of them requires more explanation to complete the activity.
  - ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
    - It fosters oral communication skills in the students and encourages them to exchange ideas with their peers. When they talk this subject with their peers, they become more knowledgeable and confident about it.
  - ❖ ***Challenges faced in implementation:***
    - Making all of the students participate and present actively in this activity is difficult.
    - More time is needed for discussion and for everyone to contribute their views.

References:

❖ https://www.ritrjpm.ac.in/images/computer-science/CS8451-Think-Pair-Share-PJT.pdf

❖ https://www.ritrjpm.ac.in/images/computer-science/14_CS8392_TPS.pdf

Signature of Faculty Member **HOD**